

Simple Interest

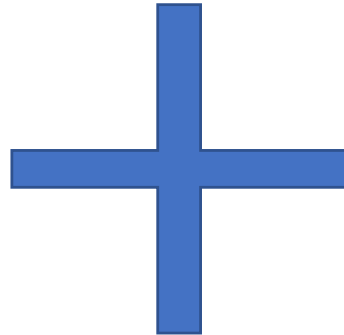


Borrowed money
or investment



Repayment or final
amount





Principal
(Borrowed
money or
investment)

Interest



Repayment or final
amount

Interest

- Amount of money paid when one borrows money or invest
- The amount of **interest (I)** paid is dependent on
 - The **principal (P)** – the initial amount of money which is borrowed or invested
 - The **rate (R)** at which the interest is charged
 - The **time (T)** for which the money is borrow.

SIMPLE Interest

$$I = PRT$$

$$P = \frac{I}{RT}$$

$$R = \frac{I}{PT}$$

$$T = \frac{I}{PR}$$

I = interest

P = principal

R = rate (%)

T = time

- How much interest would I have to pay if I borrow $\underbrace{\$4,000}_{P}$ for $\underbrace{6 \text{ years}}_T$ at an simple interest rate of $\underbrace{6\% \text{ per annum}}_R$?

$$I = PRT$$

$$= 4000 \times 6\% \times 6$$

$$= 1440$$

The interest I have to pay is \$1440

Mark borrowed $\overset{P}{\$3000}$ from his parents to buy a motorbike and he paid back a total of $\overset{P+I}{\$3,600}$ after $\overset{T}{5 \text{ years}}$ which includes simple interest on the loan. How much was the interest rate?

$$I = PRT$$

$$\text{Given } P+I = \$3600$$

$$\begin{aligned} I &= \$3600 - \$3000 \\ &= \$600 \end{aligned}$$

$$\begin{aligned} R &= \frac{I}{PT} = \frac{600}{3000 \times 5} \\ &= 0.04 \end{aligned}$$

The simple interest rate is 4% p.a.

- The interest paid on a loan was $\overset{I}{\$2,000}$. Given that the interest rate is 5% and the money was borrowed over 10 ^T years, what is the principal?

$I = 5\%$ p.a. simple [NOT Given]
ASSUMED

$$P = \frac{I}{RT} = \frac{\$2000}{5\% \times 10} = 4000$$

The principal = \$4000

What if T is less than 1 year?

Given that the interest rate is always given per annum (per year), if T is less than one year, we can calculate the interest based on the duration.

- For T in days, $I = \frac{PRT}{365}$
- For T in months, $I = \frac{PRT}{12}$
- For T in quarters (3 months), $I = \frac{PRT}{4}$
- For T in half year (6 months), $I = \frac{PRT}{2}$

$T = 75$ days

How much interest is due after 75 days for an investment of \$25 678 invested at 8% per annum?

P

R

$$I = \frac{PRT}{365} = \frac{25678 \times 8\% \times 75}{365}$$
$$= 422.104$$

The interest is \$422.10

How much is the interest rate per annum if I invested \$88 000^A in an account which yield an interest of \$289 after 3 months?
 (Give your answers to 2 decimal places) I T = 3 months

$$I = \frac{PRT}{12}$$

$$R = \frac{12 I}{P T} = \frac{12 (289)}{88000 \times 3}$$

$$= 0.0131$$

$$R = 1.31\%$$

The interest rate is 1.31%